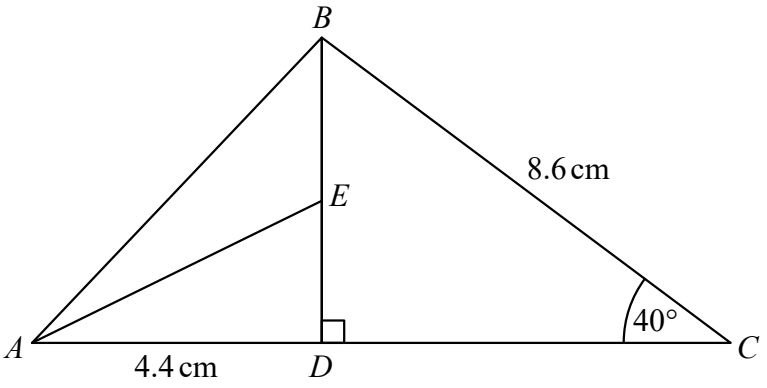
12 The diagram shows triangle *ABC*.



ADC and *DEB* are straight lines.

AD = 4.4 cm
BC = 8.6 cm
E is the midpoint of *DB*.

Angle *CDB* = 90°
Angle *DCB* = 40°

Work out the size of angle *EAD*.
Give your answer correct to 1 decimal place.
You must show all your working.

In right triangle BCD, BD = 8.6 sin40°

≈ 5.528 cm. E is midpoint, so ED ≈ 2.764 cm.

In right triangle AED, tan(angle EAD) = ED/AD

= 2.764/4.4 ≈ 0.6282.

Angle EAD ≈ 32.1° (to 1 decimal place).

4 marks total
Find BD with sine
Halve to ED
Use tangent in AED
32.1°

.....°

(Total for Question 12 is 4 marks)